

Certification Program: IBM Skillsbuild PBL- From Learner to become An AI Agent Architect

Project Title :
VocalEyes “Hear the world, see the possibilities.”

Presented By: Gen-Z AI

□ **Team Members:**

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Introduction & Motivation

Context

➤ Current Challenges:

- Limited screen readers
- Low-quality or robotic audio
- Inaccessibility of images, charts, and scanned documents
- Inability to understand visual content (charts, signs, forms)
- No modular assistive tools using modern AI
- Limited availability of affordable, real-time assistive tools

➤ Potential Solution:

- A multi-agent AI system that:
 - ✓ Accepts any document/image
 - ✓ Understands content using Gemini API
 - ✓ Converts it into natural speech using Murf API
 - ✓ Outputs real-time, accessible audio

285 MILLION

These individuals face daily challenges accessing visual content - reading documents, understanding images, and navigating online media.



Problem Statement

➤ Problem Statement:

- Visually impaired individuals struggle to independently access written or visual information in physical and digital formats.

➤ Objective:

- To design a modular, scalable AI system that empowers blind users with voice-based understanding of documents and images.

➤ Specific Goals:

- Build multi-agent interaction
- Use advanced APIs (Gemini & Murf)
- Provide regional language support (future goal)
- Ensure real-time, accessible experience



Sustainable Development Goal(SDG)

➤ Goal Chosen:

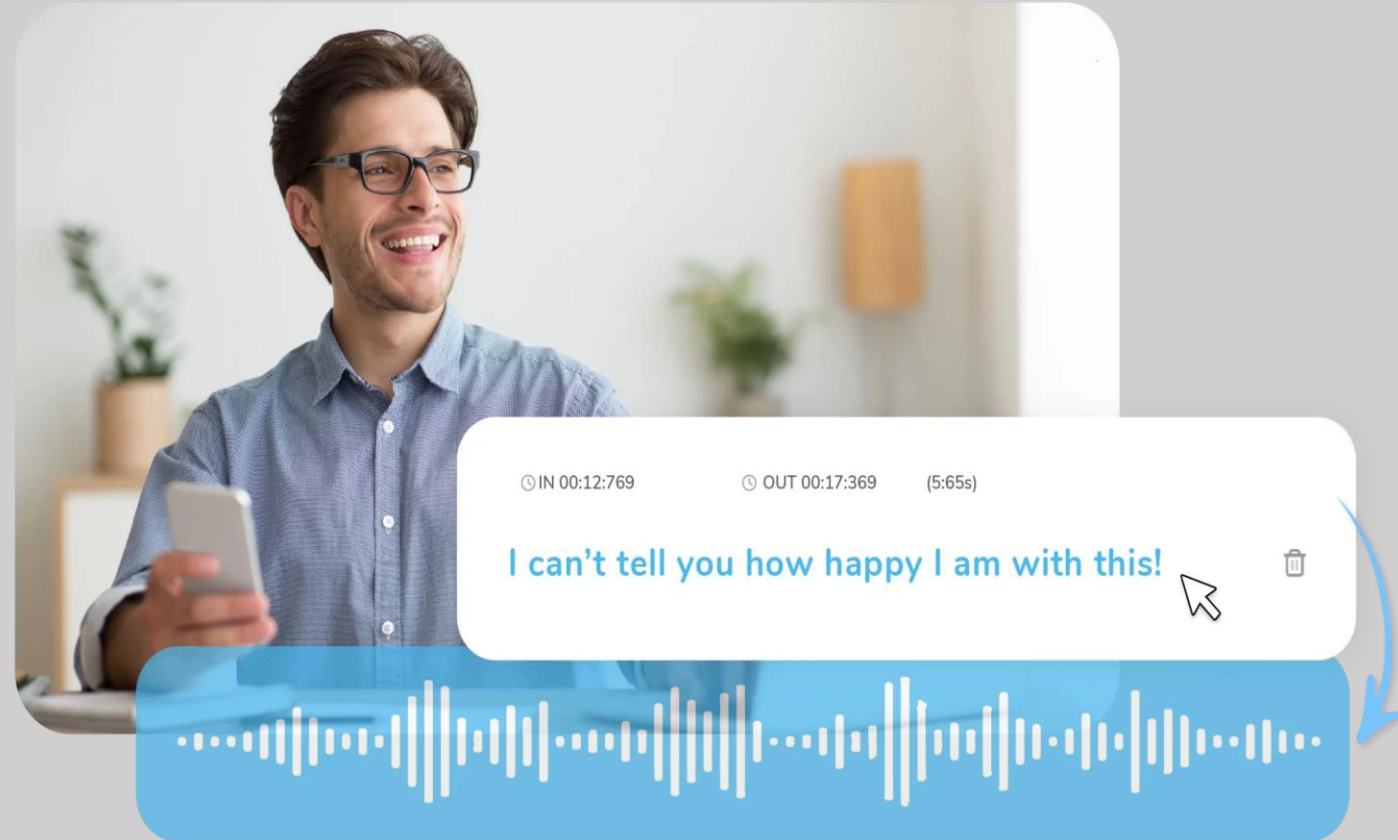
- Reduced Inequalities (Primary)
- Also supports SDG 4 (Quality Education) and
- SDG 3(Well-being)

➤ Problem statement :

- Millions of blind people face barriers in accessing information. VocalEyes reduces inequality by enabling autonomous access to any content via voice.“

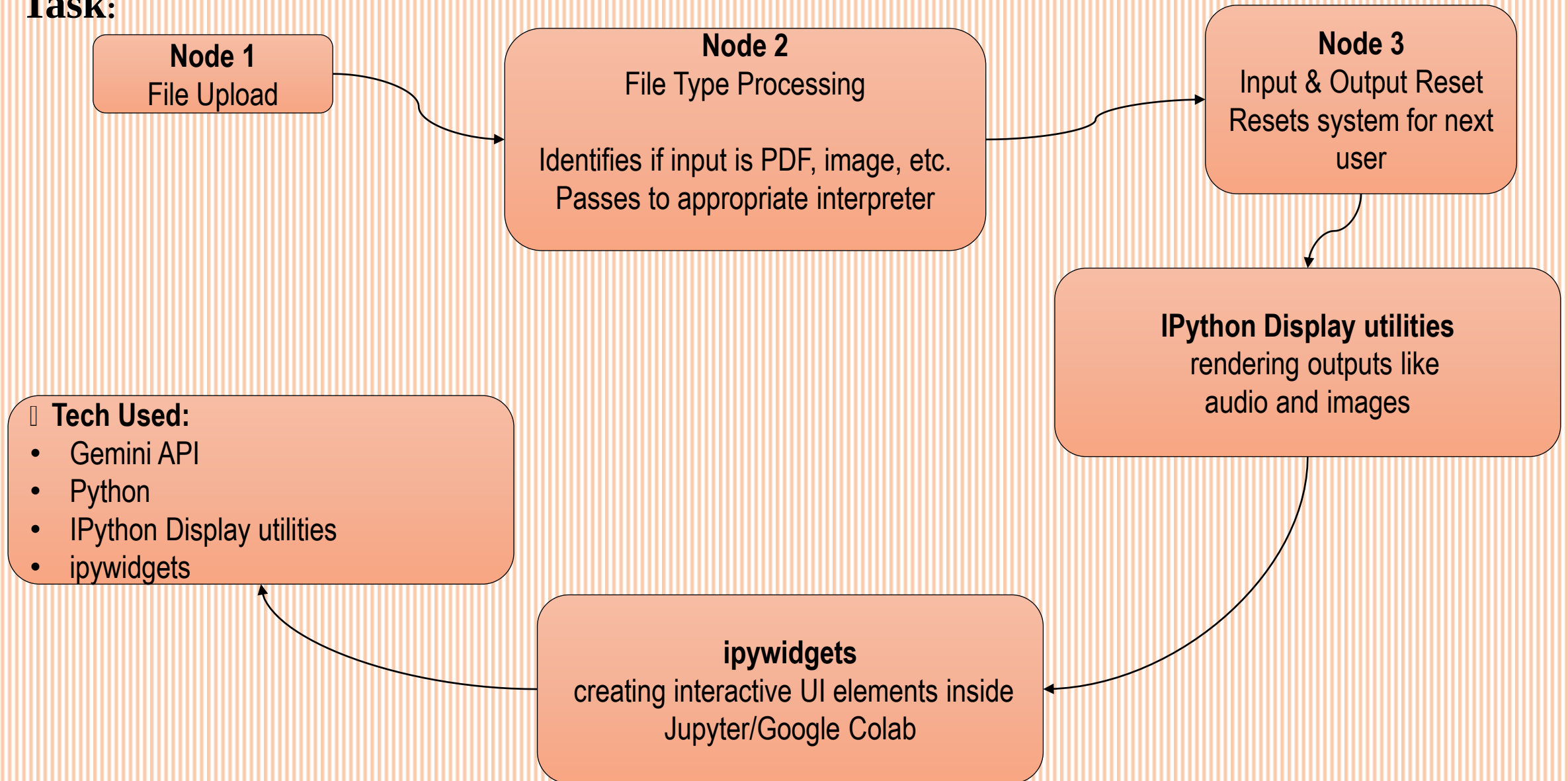
□ Rationale :

- Creating inclusive technology is critical to achieving equity in education, employment, and access to services.



Agent 1 – Document/Image Understanding

Task:





Agent 2 – Voice Response Generator

Task:

1. Text Input from Agent 1
2. Send to Murf API
3. Generate Audio URL
4. Stream/playback audio to user

☐ Output:

- Natural-sounding speech in English (expandable to regional languages)

☐ Tech Used:

- Murf API
- Python
- Web Audio streaming

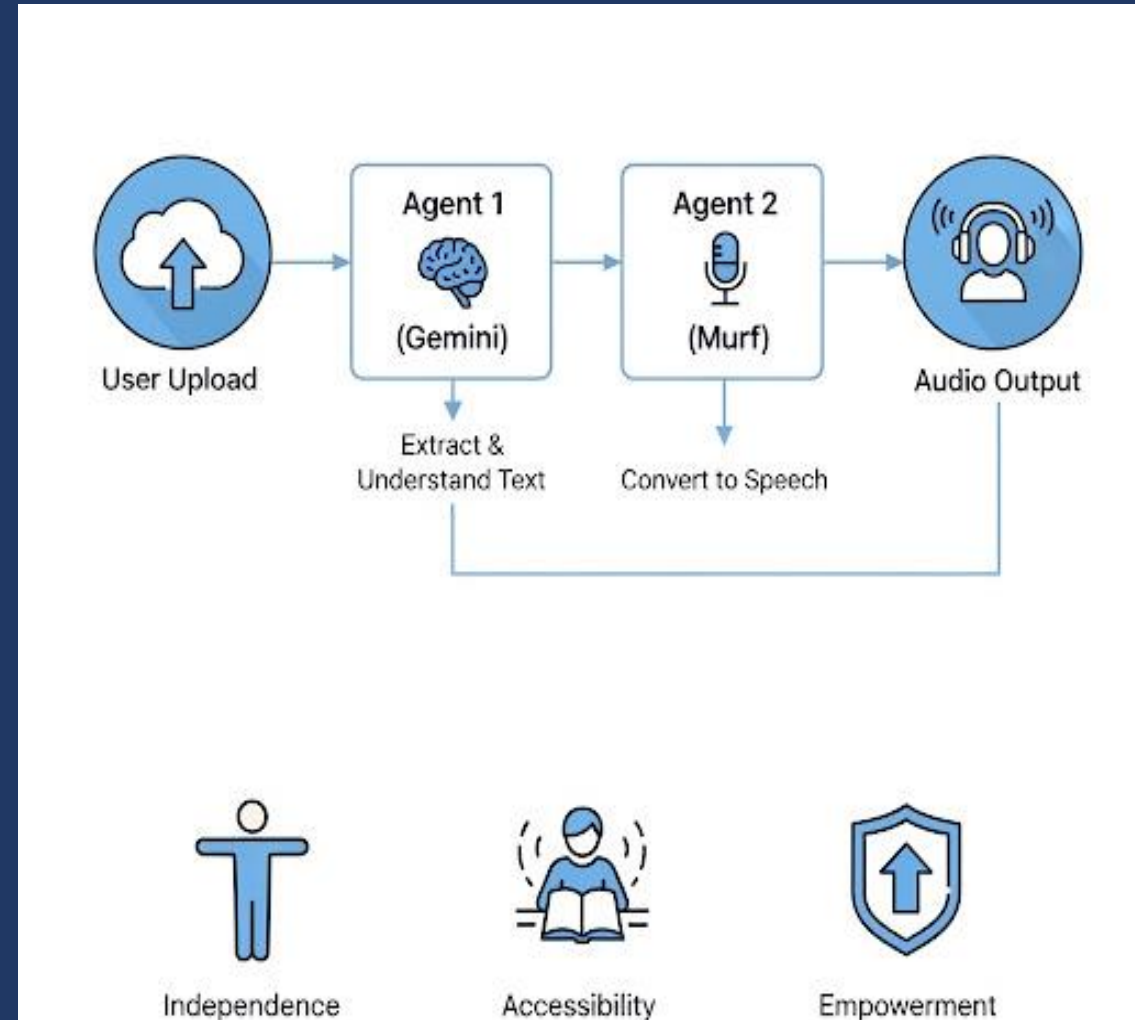
Multi-Agent Workflow Diagram

➤ Workflow Summary:

- User uploads file
- Agent 1 (Gemini) extracts/understands text
- Text passed to Agent 2
- Agent 2 uses Murf to convert to speech
- Audio is streamed to the user

□ Result:

- Independence, accessibility and empowerment





Tools & Resources

Tool / API	Usage
Gemini API	Image/doc text extraction, content analysis
Murf API	High-quality text-to-speech
Python	Backend logic, API integration
Ipywidgets	For creating interactive UI elements
IPython Display utilities	For rendering outputs like audio and images.
GitHub	Project version control

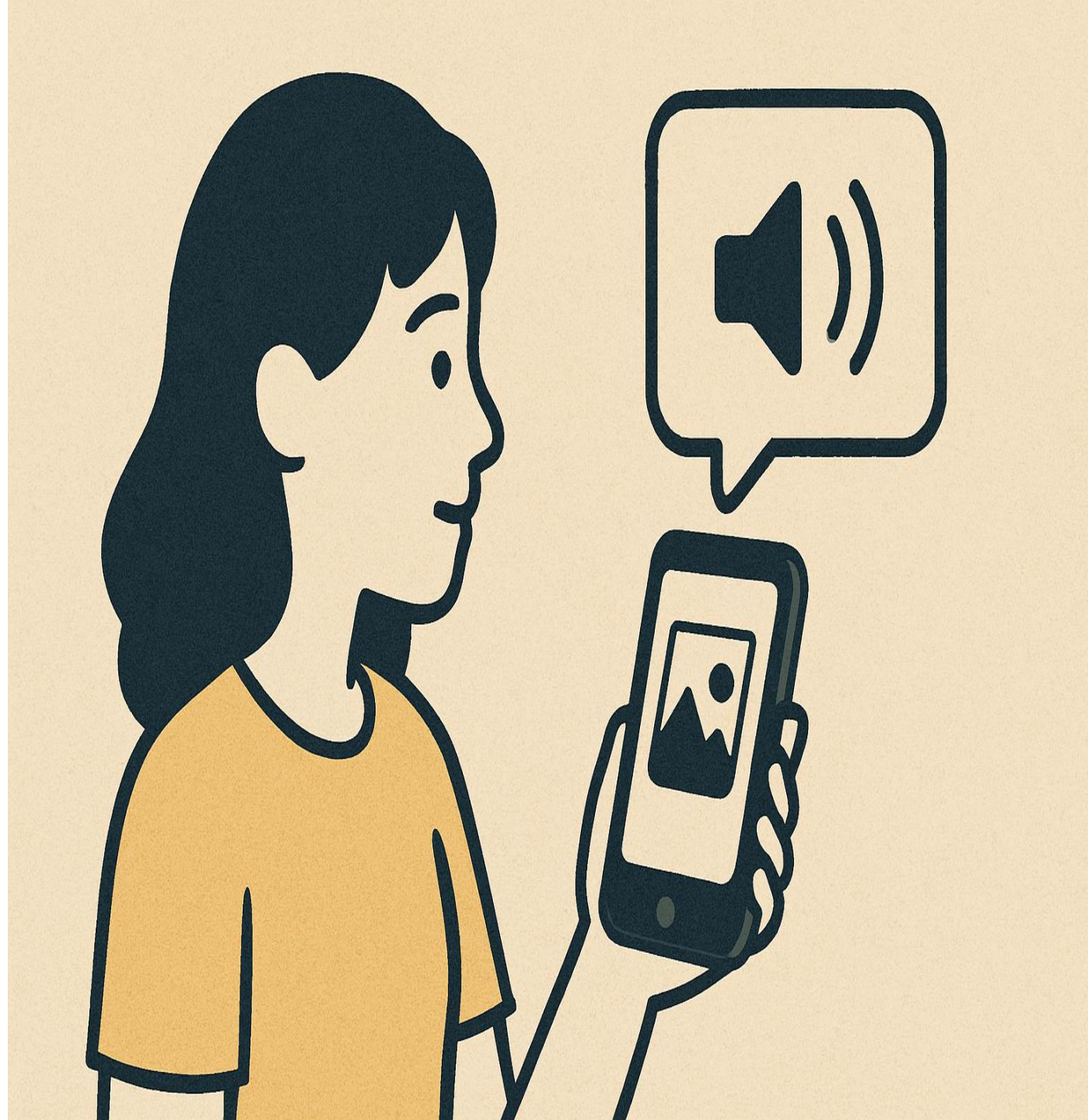


Impact & Effectiveness

- **Reach & Inclusivity:**
 - 285+ million visually impaired globally useful in education, employment, government, health
- **Impact Areas:**
 - Access to learning material
 - Reading official forms & reports
 - Understanding daily documents without help
- **AI Agent Benefits:**
 - Scalable, plug-and-play
 - Easy to extend with more APIs
 - High performance with minimal user effort

Why It Will Work

- ✓ Uses cutting-edge APIs (Gemini + Murf)
- ✓ Multi-agent = scalable, maintainable
- ✓ Simple, intuitive UI
- ✓ Real-world need with clear SDG alignment
- ✓ Cloud-based = easy deployment
- ✓ Natural voice output = high user satisfaction
- ✓ Extensible: can add Braille output or translation



Screenshots / Results

 **VocalEyes**
Hear the World, See the Possibilities

Upload a document or image → Analyze with Our AI Agent → Convert Result to Audio

Supported File Types: .txt .pdf .docx .jpg .jpeg .png .gif
Maximum Upload size: ~100 MB

📎 Upload File (1)

Analyze File

🔊 Text to Speech

■ Text Output

🔊 Audio Output

Screenshots / Results(.pdf file extraction)

ABSTRACT

The integration of Agentic AI into scientific discovery marks a new frontier in research automation. These AI systems, capable of reasoning, planning, and autonomous decision-making, are transforming how scientists perform literature review, generate hypotheses, conduct experiments, and analyse results. This survey provides a comprehensive overview of Agentic AI for scientific discovery, categorizing existing systems and tools, and highlighting recent progress across fields such as chemistry, biology, and materials science. We discuss key evaluation metrics, implementation frameworks, and commonly used datasets to offer a detailed understanding of the current state of the field. Finally, we address critical challenges, such as literature review automation, system reliability, and ethical concerns, while outlining future research directions that emphasize human-AI collaboration and enhanced system calibration.

INTRODUCTION

The rapid advancements of Large Language Models (LLMs) (Touvron et al., 2023; Anil et al., 2023; Achiam et al., 2023) have opened a new era in scientific discovery, with Agentic AI systems (Kim et al., 2024; Guo et al., 2023; Wang et al., 2024; Abramovich et al., 2024) emerging as powerful tools for automating complex research workflows. Unlike traditional AI, Agentic AI systems are designed to operate with a high degree of autonomy, allowing them to independently perform tasks such as hypothesis generation, literature review, experimental design, and data analysis. These systems have the potential to significantly accelerate scientific research, reduce costs, and expand access to advanced tools across various fields, including chemistry, biology, and materials science.

Recent efforts have demonstrated the potential of LLM-driven agents in supporting researchers with tasks such as literature reviews, experimentation, and report writing. Prominent frameworks, including LitSearch (Ajith et al., 2024), ResearchArena (Kang & Xiong, 2024), SciLitLLM (Li et al., 2024c), CiteME (Press et al., 2024), ResearchAgent (Baek et al., 2024) and Agent Laboratory (Schmidgall et al., 2025), have made strides in automating general research workflows, such as citation management, document discovery, and academic survey generation. However, these systems often lack the domain-specific focus and compliance-driven rigor essential for fields like biomedical domain, where the structured assessment of literature is critical for evidence synthesis. For example, Agent Laboratory demonstrated high success rates in data preparation, experimentation, and report writing. However, its performance dropped significantly in the literature review phase, reflecting the inherent challenges of automating structured literature reviews. Moreover, questions about system reliability, reproducibility, and ethical governance continue to pose significant hurdles.

Audio Output

▶ 0:00 / 3:47 🔊 ⋮

Screenshots / Results(.jpeg file analysis)

 Upload File (1)

Analyze FileText to Speech

Text Output

 Processing `Agentic\_AI\_Workflow.JPG` (type: image/jpeg)

The image contains two figures illustrating the application of Agentic AI in scientific discovery.

Figure 1: Agentic AI workflow for scientific discovery depicts a circular workflow diagram. The diagram shows a series of steps, each represented by a circular image containing a cartoon robot performing a task, and connected by arrows indicating the flow of the process. The steps are:

- Idea Generation & Literature Review:** Shows a robot reviewing documents.
- Research Planning & Experiment Design:** Shows a robot analyzing a graph.
- Data Preparation & Experiment Execution:** Shows a robot in a lab setting.
- Results Analysis:** Shows a robot analyzing data on a computer screen.
- Report Writing & Synthesis:** Shows a robot writing a report.
- Paper Review:** Shows a robot reviewing a document.

The circular nature emphasizes the iterative nature of the scientific process. The use of cartoon robots emphasizes the automation aspect of Agentic AI.

Figure 2: AI Agents frameworks for scientific discovery presents a tree-like diagram categorizing different AI agent frameworks across various scientific disciplines. The main branches represent scientific fields: Chemistry, Biology, Materials Science, General Science, and Machine Learning. Each branch further subdivides into specific AI agent tools and projects, listed with their names and sometimes citations. This figure provides a taxonomy of available AI tools for different scientific research areas.

In summary, the image visually explains both the process (Figure 1) and the available tools (Figure 2) of using Agentic AI in scientific research. The use of clear and concise visuals, along with text labels, makes the information easily digestible.

Audio Output

 0:00 / 2:05  

Project Links

- ❑ Google Colab Link: [Click Here](#) (Recommended)
- ❑ Github Repo Link: [Click Here](#)
- ❑ **Important:** Our project generates audio output which is not possible to showcase in a PPT. So, the generated output audio files and provided input files are uploaded [Here in Google Drive](#).
- ❑ Note: Please Get APIs in order to run the code.
Google Gemini API: <https://aistudio.google.com/apikey>
Murf API key: <https://murf.ai/api/docs/text-to-speech/overview>

Conclusion & Future Scope

□ Conclusion:

- **VocalEyes** bridges the accessibility gap for the visually impaired using intelligent AI agents.
- It's a step toward a more inclusive, voice-powered digital world.

□ Future Scope:

- Support for regional languages
- Offline version for remote use
- Integrate object recognition and scene narration
- Add Braille reader compatibility
- Government-level deployment with NGO support

Thank You!

